D 32382 **(Pages : 2)** **Name.....**

Reg. No.....

FIRST SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2022

Physics/Applied Physics

PHY 1B 01/APH 1B 01—MECHANICS–I

(2020—2022 Admissions)

Time : Two Hours Maximum : 60 Marks

The symbols used in the question paper have their usual meanings.

Section A (Short Answer Type)

Answer **all** *questions in two* **or** *three sentences, each correct answer carries a maximum of* 2 *marks.*

1. Define and explain mass.
2. Define inertial frame of reference with an example.
3. Explain the essence of Newton's first law.
4. Explain the gravitational force of a sphere.
5. Name the four fundamental forces in nature ?
6. find the tension of a "Dangling rope" at a distance x from the bottom.
7. State work-energy theorem for a conservative system.
8. Explain one example for conservation of linear momentum.
9. Show that angular momentum is conserved for a particle in central force motion.
10. Write a note on conservation of mechanical energy.
11. Give the relation between P.E and force.
12. Define normal force.

(Ceiling 20)

Turn over

Section B (Paragraph / Problem Type)

Answer **all** *questions in* **a paragraph** *of about* **half a page to one page,** *each correct answer carries a maximum of* 5 *marks.*

13. 'Ideal point masses in not essential for applying Newtons laws'. Explain with example.

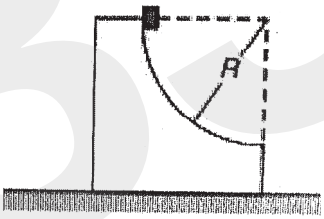
14. Write a note on Astronauts' Tug-of-War.

15. Explain Torque due to gravity.

16. A 5-kg mass moves under the influence of a force $\mathrm{F} = \left(4t^2\,\hat{i} - 3t\,\hat{j}\right)$ N, where t is the time in seconds. It starts at rest from the origin at t = 0. Find : (a) its velocity ; (b) its position ; and (c) $r \times v$, for any later time.

17. Define and explain the Principle of equivalence.

18. A small cube of mass m slides down a circular path of radius R = 10 c.m cut into a large block of mass M, as shown. M rests on a table, and both blocks move without friction. The blocks are initially at rest, and m starts from the top of the path. Find the velocity v of the cube as it leaves the block.



19. Explain work- energy theorem for a rigid body.

(Ceiling 30)

Section C (Essay Type)

Answer in about **two pages,** *any* **one** *question.*
Answer carries 10 *marks.*

20. Define and explain conical pendulum. Derive the equation for the angle the rod that makes with the vertical.

21. With figure solve the equation of simple harmonic motion.

(1 × 10 = 10 marks)